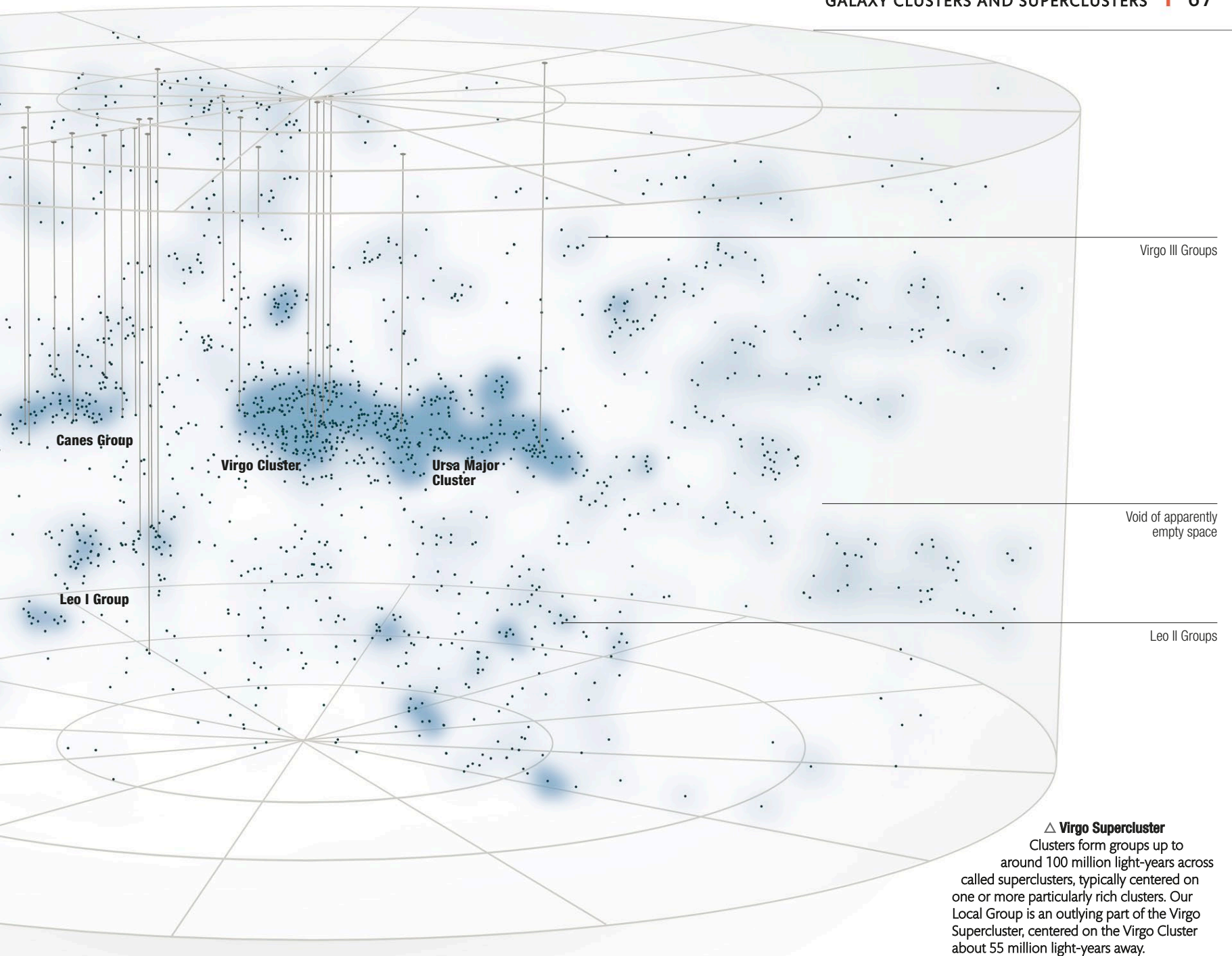




Cluster gas

Images of dense galaxy clusters taken by orbiting X-ray telescopes reveal that much of the apparently empty space between galaxies in most clusters is filled with superheated gas at temperatures of 10 million degrees or more. This hot gas is thought to originate in the cluster's individual galaxies, and to escape when it is heated up during collisions between cluster members. Hotter gas particles move faster and find it easier to escape from the gravity of their original galaxies, though not from the cluster as a whole. X-ray gas tends to accumulate at the center of a cluster over time, and is richest in the densest galaxy clusters, where it may weigh twenty times more than all the visible galaxies put together.

▷ Evolving cluster

This image of galaxy cluster IDCS J1426 combines visible, infrared, and X-ray views (in green, red, and blue respectively). It shows how the X-ray gas has largely disconnected from the visible galaxies, except where it concentrates around a recent galaxy collision.

